

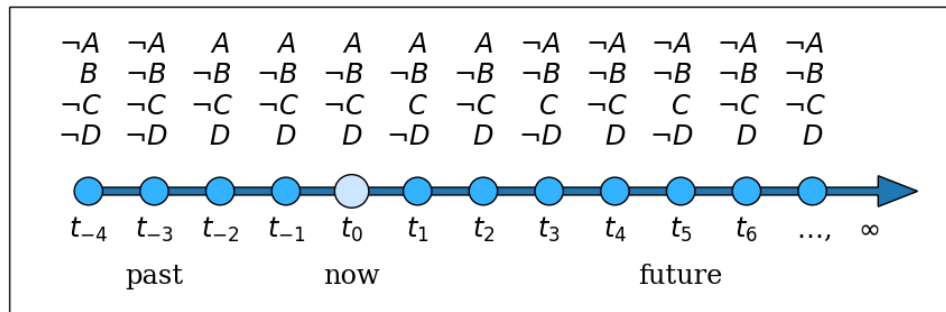
An Exercise on Logical Representations of Time

In this exercise we look at temporal representation from three different perspectives:

- Part 1 looks at how a tense logic model (an ordered sequence of propositional logic models) determines the truth of propositional formulae with tense operators.
- Part 2 is about using the representation to represent actual temporal information as expressed in English sentences.
(Some of you may prefer to do Part 2 first. In fact it may be a good idea to look at both Parts 1 and 2 and then go back and forth between the two answering the specific questions of each part.)
- Part 3 considers the use of a 1st-order temporal logic, which provides a different way of representing temporal information.

1 Tense Logic Models

Consider the Tense Logic model illustrated below. Here time starts at a specific point t_{-4} (the big bang) and goes infinitely into the future. But once t_6 is reached the same state continues infinitely into the future.¹



For each of the following formulae, determine whether it is True or False in the model. A formula should be considered True if it is true 'now', i.e. at the world state of t_0 .

1. $A \rightarrow B$
2. $A \wedge \mathbf{P}\neg A \wedge \mathbf{F}\neg A$
3. $\mathbf{P}B \wedge \mathbf{G}PB$
4. $\mathbf{H}PB$
5. $\mathbf{G}(C \vee D)$
6. $\mathbf{F}C \wedge \mathbf{F}D$
7. $\mathbf{F}(C \wedge D)$
8. $\mathbf{P}A \rightarrow \mathbf{G}(C \vee D)$

Meaning of the tense operators:

- $\mathbf{F}\phi$ ' ϕ is true at some future time.'
 $\mathbf{P}\phi$ ' ϕ is true at some past time.'
 $\mathbf{G}\phi$ ' ϕ is true at all future times.'
 $\mathbf{H}\phi$ ' ϕ is true at all past times.'

9. $\mathbf{F}A \wedge \mathbf{F}FA$
10. $\mathbf{F}FA$
11. $\mathbf{H}((A \wedge \neg B) \rightarrow D)$
12. $\mathbf{G}(B \rightarrow D)$

¹Does it make sense to say that time passes on when nothing changes? In this general kind of model this is possible, since a state could be exactly the same as the previous state; but we could place restrictions on allowed models to take into account other conditions we may want to capture, regarding the nature of time. But in current example where, we only specify four propositions, we may suppose that although A , B , C and D may stay the same from one state to the next, other facts about the world could be changing.

2 Representation in Tense Logic

(a) Translate each of the following sentences into tense logic:

1. I have never eaten a snail.
2. I will only eat a haggis if I am in Scotland.
3. If you have not bought a ticket you will not be able to go to the concert.
4. I have eaten broccoli and chocolate but never at the same time.
5. I will visit Paris and then after that I will visit Rome.
6. Leeds was always beautiful, except when it was raining.
7. I shall not go out before the baby sitter arrives.
8. If I win the lottery I will buy a helicopter.
9. I've not seen her since before I first met you.
10. You will always be lucky or unlucky but in either case your luck will change.

(b) Give a natural language sentence corresponding to the tense logic formula

$$\mathbf{P}\chi \rightarrow \mathbf{G}(\mathbf{F}\chi \vee \neg\xi),$$

where χ represents the proposition ‘She eats chocolate’ and ξ represents the proposition ‘She is alive’.

(c) In tense logic *define* a new construct $\mathbf{A}\phi$, which means there is no time (past, present or future) when ϕ is false — i.e. ϕ is always true. Your definition should be of the form $\mathbf{A}\phi \equiv_{def} \Phi(\phi)$, where $\Phi(\phi)$ should be replaced by some formula containing only the standard propositional operators, the standard tense operators and one or more occurrences of ϕ .

3 A 1st-Order Temporal Representation Language

Let $\mathcal{TL}$ be a 1st-order temporal language containing time variables, $t_1, \dots, t_n$, a strict temporal ordering relation $<$, a constant ‘now’ denoting the current time and a construct $\text{HoldsAt}(\phi, t)$ meaning that ϕ is true at time t .

1. Use $\mathcal{TL}$ to give a formula which means the same as the tense logic formula $\mathbf{F}A$.
2. Use $\mathcal{TL}$ to represent the sentence ‘It will not rain before it gets cloudy.’
3. Explain concisely the meaning of the $\mathcal{TL}$ formula

$$\forall t_1 \forall t_2 [(t_1 < t_2) \rightarrow \exists t_3 [(t_1 < t_3) \wedge (t_3 < t_2)]].$$

4. Is propositional tense logic more or less *expressive* than $\mathcal{TL}$?

4 An axiomatic proof within Prior's tense logic

We'll prove that $\mathbf{PP}\phi \rightarrow \mathbf{P}\phi$ is a valid theorem in the Prior's tense logic system presented in the lecture slides, i.e. that it can be derived from Prior's axioms and rules presented in the lecture slides.

We will first prove the following lemma:

Lemma 4.1. *The following is a theorem in Prior's system: $\mathbf{PG}\phi \rightarrow \phi$:*

Proof of Lemma 4.1: $\phi \rightarrow \mathbf{HF}\phi$ is an axiom in Prior's tense logic (Axiom 3 from the lecture slides). Therefore, by substitution rule we have that $\neg\phi \rightarrow \mathbf{HF}\neg\phi$ is also a theorem. Therefore we have that $\neg\phi \rightarrow \neg\mathbf{P}\neg\neg\mathbf{G}\neg\neg\phi$ is a theorem, using the interdefinability of $\mathbf{H}$ and $\mathbf{P}$, and of $\mathbf{F}$ and $\mathbf{G}$. Thus, by propositional logic we have that $\neg\phi \rightarrow \neg\mathbf{PG}\phi$ is a theorem. By contraposition we can conclude that $\mathbf{PG}\phi \rightarrow \phi$ is a theorem.

Theorem 4.2. *$\mathbf{PP}\phi \rightarrow \mathbf{P}\phi$ is a valid theorem in Prior's logic.*

Proof of Theorem 4.2: $\mathbf{P}\phi \rightarrow \mathbf{GP}\phi$ is an axiom of Prior's tense logic (Axiom 5 from slides). Using the tense logic rules we get that $\mathbf{PP}\phi \rightarrow \mathbf{PGP}\phi$ is also a theorem. Moreover, by Lemma 4.1 and substitution rule, we have that $\mathbf{PGP}\phi \rightarrow \mathbf{P}\phi$ is a theorem. Therefore, by concatenation rule we get that $\mathbf{PP}\phi \rightarrow \mathbf{P}\phi$ is a valid theorem in Prior's logic.

Answers to parts 1–3 can be found on the next page. However, you are strongly advised to spend some time trying to figure out the answers yourself before looking at the given answers.

Answers

1. Tense Logic Models

- 1.1. False, since A is True and B is False at the present world, t_0 .
- 1.2. True. A is true now but was False at some time in the past and will also be False at some time in the future.
- 1.3. True. B was true at some point in the past (t_{-4}) and it will always be the case that B was true in the past.
- 1.4. False. For, $\mathbf{HP}B$ to be true we would need to have $\mathbf{P}B$ true at all points before t_0 ; but at t_{-4} there is no earlier time point, so $\mathbf{P}B$ is not true at t_{-4} , which means that $\mathbf{HP}B$ is not true at t_0 .
(In this model, t_{-4} is the first point of time — perhaps the time of the *big bang*.)
- 1.5. True. Yes at all points after t_0 , either C or D is true.
- 1.6. True. There is a time after t_0 when C is True (e.g. t_1) and also a time after t_0 when D is True (e.g. $2t_2$).
- 1.7. False. There is no time after t_0 , when both C and D are true.
- 1.8. True. A was true at some time in the past so in order for $\mathbf{P}A \rightarrow \mathbf{G}(C \vee D)$ to be true, which is indeed the case (as we saw in case 1.5).
- 1.9. True. There is a time in the future where A is true (either of t_1 and t_2) and there is also a time in the future where $\mathbf{F}A$ is true (only at t_1).
- 1.10. False. t_1 and t_2 are the only future points where A is true. SO there is no point which has a point in its future that has a point in its future, where A is True. (If time occurs in discrete steps then by nesting the $\mathbf{F}$ operator we can refer to points further and further in the future in relation to each other. (If time were *dense* (meaning that between every two points there is another point) this would not work. In dense time we would always have $\mathbf{F}\phi \rightarrow \mathbf{FF}\phi$ (or even $\mathbf{F}\phi \rightarrow \mathbf{FFFFFF}\phi$).
- 1.11. False. Yes, because at every point before t_0 we see that in all cases where A is true but B false, we also have D true.
- 1.12. True. This is true in a trivial way. Since B is always False in the future then according to the (truth-functional) interpretation of $\rightarrow$, the formula $B \rightarrow D$ will also be true at all points in the future.

More answers on next page.

2. Representation in Tense Logic

2.1. $\neg \mathbf{P} \text{ leatSnail}$ (or $\mathbf{H} \neg \text{leatSnail}$)

2.2. $\mathbf{G}(\text{leatHaggis} \rightarrow \text{linScotland})$

Note. We are representing what I take to be the usual interpretation, which is that the precondition of being in Scotland refers to any points in the future when I might eat Haggis. One could perhaps argue that there is another interpretation as equivalent to “If I am now in Scotland then at some point in the future I will eat haggis”; but this does not seem to be a natural interpretation.

2.3. $\neg \mathbf{P} \text{BoughtTicket} \rightarrow \neg \mathbf{F} \text{GoToConcert}$

2.4. $\mathbf{P} \text{EatBroccoli} \wedge \mathbf{P} \text{EatChocolate} \wedge \neg \mathbf{P}(\text{EatBroccoli} \wedge \text{EatChocolate})$

2.5. $\mathbf{F}(\text{IvisitParis} \wedge \mathbf{F} \text{IvisitRome})$

Note. The nesting of the $\mathbf{F}$ operator is used to represent the condition that the visit to Rome is in the future relative to the visit to Paris. This means that there is a point in the future where it is true that you are visiting Paris and at that point your visit to Rome is still in the future.

2.6. $\mathbf{H}(\text{LeedsBeautiful} \vee \text{Raining})$

2.7. $\mathbf{G}(\text{IGoOut} \rightarrow \mathbf{P} \text{BabySitterArrive})$??

This captures some of the meaning, but is arguably not a completely satisfactory representation. It would also be true if the baby sitter had arrived on a previous occasion but is not here now, in which case I could go out without waiting for them to arrive again. This is probably not what is intended. We could potentially add $\neg \text{BabySitterArrive}$ and $\neg \mathbf{P} \text{BabySitterArrive}$, which would then mean that I could only go out if the baby sitter arrives at some point in the future. That may be better, but it also implies that I never had a baby sitter arrive on any previous occasion, which doesn't seem to be implied by the English sentence. To get a more faithful representation of the sentence we would need to use a first-order temporal logic, which is more expressive than the Tense Logic representation.

2.8. $\mathbf{G}(\text{WinLottery} \rightarrow \mathbf{F} \text{BuyHelicopter})$

The sentence seems to mean that if at any time I win the lottery, then, at some point after that, I will win buy a helicopter. This requires nesting the $\mathbf{F}$ operator within the $\mathbf{G}$.

2.9. $\mathbf{P}(\text{IMetYou} \wedge \neg \mathbf{P} \text{IMetYou} \wedge \mathbf{P}(\text{ISawHer} \wedge \mathbf{G} \text{ISawHer}))$

Quite complex but seems to accurately capture the relative timings expressed in the English sentence.

2.10. $\mathbf{G}(\text{Lucky} \vee \text{Unlucky}) \wedge \mathbf{G}(\text{Lucky} \rightarrow \mathbf{F} \text{Unlucky}) \wedge \mathbf{G}(\text{Unlucky} \rightarrow \mathbf{F} \text{Lucky})$

This seems to capture the alternating change between being lucky and unlucky continuing into the future. However, one might say it is not quite right because it allows the possibility that time just stops; and this could happen even at the very next time point, so I would never get to be either lucky or unlucky. To ensure time does go on forever we could either add a general axiom such as $\mathbf{GF}(\phi \vee \neg \phi)$; or just one specific case would do, such as $\mathbf{GFLucky}$.

More answers on next page.

3. A First-Order Temporal Representation Language

3.1. $\exists t[\text{HoldsAt}(A, t) \wedge (\text{now} < t)]$

3.2. $\forall t[(t > \text{now} \wedge \text{HoldsAt}(\text{Raining}, t)) \rightarrow \exists t'[(\text{now} < t') \wedge (t' < t) \wedge \text{HoldsAt}(\text{Cloudy}, t')]]$

3.3. The formula insures that time is *dense*, meaning that for any two time points where the first is before the second, there is another time point that is after the first one and before the second.

3.4. $\mathcal{TL}$ is more expressive than Tense Logic.

This is actually illustrated by the answer to 3.2. This formula directly constrains a time point in relation to two other time points. We cannot always do this with temporal logic (although for certain cases it is possible).